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Modeling Self-Discharge in Li/S Batteries Through Electrochemical Anode Reactions: A Theoretical Perspective

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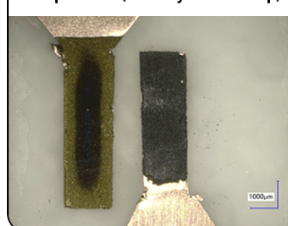
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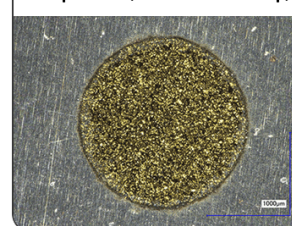
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Modeling Self-Discharge in Li/S Batteries Through Electrochemical Anode Reactions: A Theoretical Perspective

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The growing demand for high-energy-density batteries has renewed interest in lithium–sulfur (Li/S) systems, which offer significant advantages but suffer from severe self-discharge during rest. While prior studies attribute this degradation to chemical parasitic reactions or polysulfide shuttling, they overlook the inherently electrochemical nature of anode-side processes. In this work, a 1D physics-based model of a Li/S battery was developed to explicitly incorporate lithium-metal oxidation and the stepwise electrochemical reduction of polysulfides at the anode. Using COMSOL Multiphysics, galvanostatic discharge followed by open-circuit rest under two conditions was analyzed - with and without parasitic anode reactions. The results show that when anode reactions are excluded, species concentrations stabilize, and voltage remains constant during rest. In contrast, the inclusion of electrochemical reactions at the anode results in slow voltage decay and continued reduction of sulfur species, including the growth of lower order polysulfides and $\text{Li}_2\text{S}_{(s)}$ precipitation. These findings reproduce key trends observed in experimental data. This study demonstrates that electrochemical anode reactions are central to self-discharge. Resolving these mechanisms through Faradaic kinetics yields a mechanistic foundation for predictive models and underscores the need to reconsider self-discharge as an electrochemically driven phenomenon.

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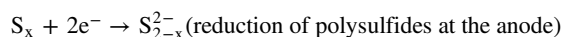
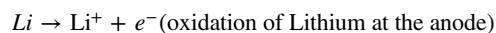
Demand for electric vehicles (EVs) and other forms of e-mobility has increased significantly in recent years all over the world. To meet the performance expectations of the current expanding market, there is a critical need for high-energy-density, lightweight, and cost-effective battery systems that can deliver long driving ranges, fast charging, and reliable performance over extended periods of rest.

Among the emerging candidates for next-generation batteries, lithium-sulfur (Li/S) batteries are especially promising due to their high theoretical energy density (2,600 Wh/kg_{active material}), low material cost, and environmental compatibility. The high specific capacity of sulfur (1,672 mAh g⁻¹) offers a compelling path to reduce battery pack weight and increase range which are key challenges in EV adoption, for example.

Li/S batteries operate by using lithium metal as the anode and elemental sulfur (or a sulfur-blend) as the cathode, connected through an ether-based electrolyte. During discharge, lithium is oxidized at the anode, while sulfur is reduced at the cathode through a series of multistep reactions that produce soluble lithium polysulfide intermediates (Li_2S_x , where $4 \leq x \leq 8$), ultimately forming insoluble Li_2S precipitate.¹ Despite intensive research on sulfur cathode engineering and polysulfide confinement strategies, the practical viability of Li/S batteries is still hindered by limitations on the anode side. Notably, the lithium metal anode is highly reactive and suffers from severe degradation, including dendritic growth, SEI instability, and parasitic reactions with soluble polysulfides.

While the polysulfide shuttle effect is widely recognized (and often misnomered or mischaracterized) for causing active material loss and coulombic inefficiency during cycling, a subtler but equally important phenomenon is self-discharge under open-circuit conditions or storage.

Self-discharge occurs in a Li/S battery under open circuit conditions because lithium metal is oxidized due to reduction of dissolved sulfur on the lithium metal. Subsequently, the polysulfide species generated due to the reduction of dissolved sulfur on the lithium metal is also reduced to a lower order polysulfide with continued oxidation of the lithium metal, as follows:



In a Li/S battery system, lithium serves as the reducing agent ($\text{Li}^0 = -3.05 \text{ V}$), and dissolved sulfur and high order polysulfide species act as oxidants, driving a large driving force and forming lower-order polysulfide species and eventually S^{2-} followed by precipitation of insoluble $\text{Li}_2\text{S}_{(s)}$ (due to the low solubility), as sketched in Fig. 1.

This process leads to passivation of the lithium surface, loss of active lithium inventory, and self-discharge during open-circuit conditions. Despite the extensive modeling of cathodic sulfur reactions and the shuttle effect, to date, no mathematical model has captured the electrochemical and chemical reduction reactions occurring at the lithium metal surface during rest conditions. However, various modeling works have been done in the Li/S field that discuss self-discharge.

Kumaresan et al.¹ presented one of the first physics-based 1D models of a Li/S cell. The authors included multistep electrochemical sulfur reduction, multicomponent transport, and precipitation/dissolution reactions within the porous cathode. Their work accurately captured the two-stage discharge profile and identified the critical roles of species solubility and solid phase formation in capacity fade. However, the anode was treated as a point and, implicitly, a boundary with lithium oxidation as the only reaction considered, without accounting for polysulfide transport to the anode or any redox activity occurring there. Many other relevant models have followed this work.^{2–10}

Other modeling efforts have implicitly or explicitly addressed the influence of anode-side processes in lithium-sulfur cells, though few incorporate mechanistic electrochemical treatments. Ghaznavi et al. conducted a three-part sensitivity analysis on the foundational Kumaresan model² finding that the model's inability to capture charging behavior and voltage fade under rest conditions likely reflects the omission of critical interfacial reactions.^{11–13} Neidhardt et al. and Fronczek et al. developed impedance-informed models which showed dynamic resistance contributions linked to polysulfide transport between electrodes, with Fronczek further noting that low polysulfide concentrations during charge result in lower electrolyte conductivity and increased overpotentials.^{14,15} Hofmann et al. highlighted irreversible transport of sulfur species to the anode, leading

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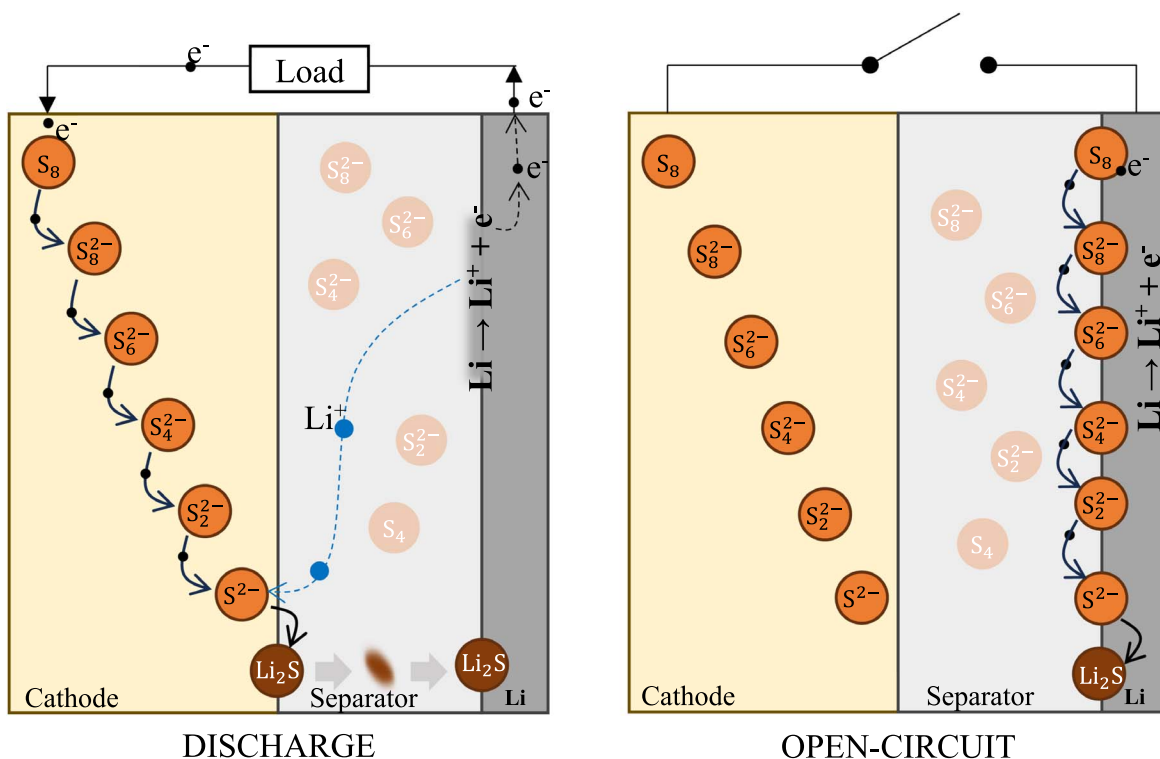


Figure 1. Sketch showing the reactions during discharge (left) and the potential reactions during open-circuit, with oxidation of Lithium disk and the the reduction of polysulfides at the anode surface (right).

to precipitation of Li_2S at the lithium surface and corresponding voltage suppression over cycling.¹⁶ Danner et al. emphasized the need for improved modeling of transport and precipitation effects within cathodes but also acknowledged that interfacial processes at the anode contribute to overall capacity loss.¹⁷ Zhang et al. demonstrated that accurate voltage prediction requires correction of electrolyte resistance and anode-side precipitation kinetics, while Moy et al. provided direct experimental evidence linking shuttle current decay to polysulfide reduction at the lithium surface.^{18,19}

Marinescu et al. and Yoo et al. investigated the voltage asymmetries and numerical instabilities tied to precipitation schemes that may be rooted in unmodeled anode-side reactions.^{20,21} Ren et al. and Shim et al. independently demonstrated that lithium nitrate alters discharge behavior by forming irreversible surface films at the anode, inhibiting further polysulfide reduction and prompting non-linear shifts in voltage profiles.^{22,23} Parke et al. introduced a reduced-order model that did not resolve species at the anode but recognized the need for such extensions, while Thangavel et al. suggested alternative reduction pathways that cannot be explained by cathode processes alone, implying overlooked interfacial chemistry at the lithium surface.^{24–26} Collectively, these studies underscore that degradation mechanisms, voltage hysteresis, and self-discharge cannot be fully understood without a mechanistic treatment of anode-side electrochemical kinetics.

Mistry and Mukherjee²⁷ advanced a rigorous physicochemical framework to describe the polysulfide shuttle, interpreting it as a two-step process involving the transport of sulfur species to the lithium anode, followed by their spontaneous reduction at the metal surface. Their model emphasized that these anode-side redox reactions persist even under open-circuit conditions, playing a dominant role in the self-discharge behavior of Li/S systems. These processes were characterized by using a chemical overpotential (η_{ch}), which governs the rate of spontaneous redox reactions at the anode surface.

Based on this foundation, Guobin Wen et al.²⁸ developed a 1D continuum model of a Li/S cell, in which self-discharge was incorporated through a lumped chemical overpotential term at the lithium metal anode. This approach enabled fitting of the apparent

rate of parasitic polysulfide reduction during rest periods and successfully reproduced experimental trends in voltage decay and capacity loss. However, this formulation treats the anode-side reactions as purely chemical, without resolving the stepwise reduction of polysulfides.

Shen et al.²⁹ conducted a systematic experimental study on the self-discharge behavior of Li/S batteries, demonstrating that lithium polysulfides, once formed, continue to diffuse toward the lithium metal anode and undergo spontaneous reduction - even under open-circuit conditions. The extent of degradation was shown to depend strongly on factors such as the depth of discharge, the concentration of polysulfides in the electrolyte, and the degree of anode protection.

In a comprehensive review, Shen et al.³⁰ highlighted the critical role of anode-side reactions in Li/S batteries. They described how long-chain polysulfides diffuse toward the lithium metal anode and undergo spontaneous reduction to insoluble species such as Li_2S_2 and Li_2S , even under open-circuit conditions. These reactions form passivating layers that contribute to both reversible and irreversible capacity loss.

Knap et al.³¹ developed a data-driven self-discharge model for Li/S batteries, grounded in direct shuttle current measurements. Their results confirmed that the dominant contributor to self-discharge is the polysulfide shuttle current, which remains active even during open-circuit rest and dynamically contributes to capacity loss. Their model treats self-discharge as a phenomenological process, without resolving the underlying electrochemical kinetics.

Moy et al.¹⁸ developed a direct experimental method for quantifying polysulfide shuttle currents by holding the cell at constant voltage and measuring the steady-state current. Their work demonstrated that higher-order polysulfides generated at the cathode diffuse to the lithium anode, driving a persistent parasitic current even at open circuit potential (OCV). Their framework treats the reactions at the lithium surface chemically, without resolving the interfacial electron-transfer kinetics.

Mikhaylik and Akridge³² provided one of the earliest quantitative models linking the polysulfide shuttle with multiple performance

Table I. Domain dimensions used in the model.

Region, m	Cathode	Separator	Anode
Thickness	$L_c = 41 \mu\text{m}$	$L_s = 9 \mu\text{m}$	0 (Boundary)

metrics in Li/S batteries. They introduced a shuttle constant to describe the reduction of high-order polysulfides at the lithium metal surface and showed that this process leads to exponential decay in capacity during storage. Their model established an essential connection between the shuttle and lithium corrosion but did not mechanistically represent the sequential anode-side reduction cascade.

Chen et al.⁵ provided a comprehensive overview of simulation approaches for Li/S batteries. While most existing models have emphasized cathode-side phenomena, computational efforts targeting the lithium metal anode remain relatively limited. The anode is mainly studied through dendrite suppression, SEI formation, and ionic transport, but the role of anode-side reductions of polysulfides, particularly under open-circuit conditions, has not been fully modeled.

Cornish and Marinescu⁹ developed a 1D electrochemical model of Li/S cells with an emphasis on experimental validation. Their framework included a simplified chemical reduction of elemental sulfur at the anode surface, governed by a temperature-dependent shuttle constant. While this represents an important step toward including anode-side chemistry, the model treats the reaction as a single-step process and does not capture the sequential reduction of polysulfides.

In addition to these modeling papers, recent experimental and mesoscale simulation studies have provided further insight into anode-side self-discharge behavior. Hao et al.³³ used mesoscale simulations to show that soluble polysulfides spontaneously reduce on the lithium surface even under open-circuit conditions, leading to Li_2S deposition and anode passivation over time. Liu and Mukherjee³⁴ demonstrated that the evolution of surface films depends on local microstructure and electrolyte properties, further supporting the need to resolve interfacial reactions at the lithium anode. More recently, Guo et al.³⁵ reviewed the role of lithium corrosion and parasitic reactions under lean electrolyte conditions, emphasizing that spontaneous redox reactions between lithium and dissolved sulfur species persist during rest and are key drivers of shelf-life degradation. These studies reinforce the importance of capturing electrochemical and chemical kinetics at the lithium surface and motivate the development of more mechanistic models for self-discharge.

Despite the extensive literature on lithium-sulfur self-discharge, an opportunity exists to refine the understanding of anode-side reactions. Prior models have characterized the interactions between lithium metal and polysulfides using frameworks like “chemical overpotential” (η_{ch}), which relate reaction rates at the anode to exchange current densities in a mathematically similar way to electrochemical formulations. The present work builds upon this foundation by explicitly modeling these processes as electrochemical half-reactions with Faradaic currents and charge transfer at the lithium metal surface. Rather than treating lithium oxidation and polysulfide reduction as separate phenomena, this approach implements them within a unified electrochemical framework that accounts for reaction kinetics across both electrodes. This methodology enables more precise modeling of the coupled processes driving voltage decay during rest periods and provides insights into developing effective strategies to mitigate self-discharge in Li/S batteries.

Model Description

Geometry.—The simulated cell corresponds to a one-dimensional geometry with three distinct regions (two domains and a boundary): the cathode (domain), the separator (domain), and the

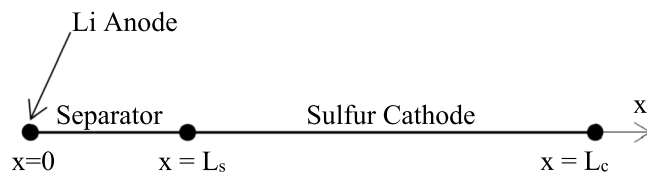


Figure 2. Sketch showing the 1D geometry representing the anode domain/boundary, the separator domain, and the sulfur-based cathode (indexed with letter “ m ” in the equations).

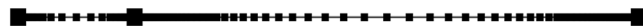


Figure 3. The distributed mesh used for the model.

lithium metal anode (boundary), as seen in Fig. 2. The geometry and the model were built in COMSOL Multiphysics v.6.3.

The thicknesses of the different regions are taken from Ref. 2 and given in Table I.

Geometry meshing.—A distributed one-dimensional mesh was used to discretize the model domains. The mesh density was specifically refined near the interfaces between the electrodes and the separator, where steep gradients in species concentrations and electric potentials are expected due to electrochemical reactions, strong diffusion limitations, and precipitation/dissolution of solid species.

A higher mesh density was chosen near the cathode-separator and separator-anode boundaries (Fig. 3) to ensure numerical stability and to resolve steep gradients in polysulfide species concentrations, electrolyte potential, and lithium-ion distribution.

The regions away from the interfaces were coarsened using an exponential element growth rate to reduce computational cost while maintaining accuracy where it is most critical. The mesh settings can be seen in Table II.

Governing equations.—The governing equations include equations for the following regions $m \in \{\text{an, sep, cat}\}$, species $i \in \{\text{Li}^+, \text{S}_{8(\text{l})}, \text{S}_{8(\text{s})}^{2-}, \text{S}_6^{2-}, \text{S}_4^{2-}, \text{S}_2^{2-}, \text{S}^{2-}, \text{and } \text{A}^-\}$ reactions: $k = k_{\text{cat}} = k_{\text{sep}} \in \{2, 3, 4, 5, 6\}$ (when no anode reactions) and $k = k_{\text{an}} \in \{1, 7, 8, 9, 10, 11\}$ (when including anode reactions) and solid precipitates: $j \in \{\text{S}_{8(\text{s})}, \text{Li}_2\text{S}_{(\text{s})}\}$.

Mass conservation.—Mass conservation for each species i within domain m (m corresponds here to the separator and cathode region for the mass conservation) is governed by the following general equation, assuming diluted species theory:

$$\epsilon_m \frac{\partial c_{i,m}}{\partial t} + \nabla \cdot \mathbf{N}_{i,m} = r_{i,m} + R_{p,j,m} - \frac{\partial \epsilon_m}{\partial t} c_{i,m} \quad [1]$$

where ϵ_m is the porosity of each domain m , $c_{i,m}$ the concentration of species i (Li^+ , $\text{S}_{8(\text{l})}$, $\text{S}_{8(\text{s})}^{2-}$, S_6^{2-} , S_4^{2-} , S_2^{2-} , S^{2-} , and A^- - Table III) in mol m^{-3} , $\mathbf{N}_{i,m}$ is the species flux in each domain m , in $\text{mol m}^{-2} \text{s}^{-1}$, ∇ is the del operator for 1D-domain (i.e., $\nabla = \partial/\partial x$), $R_{p,j,m}$ is the source or sink term in $\text{mol}/(\text{m}^3 \cdot \text{s})$ due to precipitation/dissolution of solid species j (assuming $\text{S}_{8(\text{s})}$ and $\text{Li}_2\text{S}_{(\text{s})}$ to be the only lithium polysulfide precipitates) in each domain m and $r_{i,m}$ is the source or sink term due to homogeneous electrochemical reactions involving species in each domain m , in $\text{mol m}^{-3} \text{s}^{-1}$. Here, A^- denotes the counterion of the lithium salt in the electrolyte, such as bis(trifluoromethanesulfonyl)imide (TFSI^-), which ensures electroneutrality in solution.

The species flux $\mathbf{N}_{i,m}$ is described by the Nernst-Planck equation:

$$\mathbf{N}_{i,m} = -D_{i,m}^{\text{eff}} \nabla c_{i,m} + z_i D_{i,m}^{\text{eff}} \frac{F}{RT} c_{i,m} \nabla \phi_{l,m} \quad [2]$$

Table II. Meshing parameters.

Parameter	Value				
Global Maximum Element Size	1×10^{-6} m				
Global Minimum Element Size	3.75×10^{-9} m				
Maximum Element Growth Rate	1.2				
Curvature Factor	0.25				
Predefined Size	Extra Fine				
Region	No. of Elements	Element Ratio	Growth Rate	Distribution	
Cathode (Domain 2)	100	20,000	Exponential	Symmetric	
Separator (Domain 1)	50	30,000	Exponential	Symmetric	

Table III. Species index i.

<i>i</i>	1	2	3	4	5	6	7	8
Species	Li^+	$S_{8(l)}$	S_8^{2-}	S_6^{2-}	S_4^{2-}	S_2^{2-}	S^{2-}	A^-
z_i	+1	0	-2	-2	-2	-2	-2	-1

where $D_{i,m}^{eff}$ is the effective diffusion coefficient of species i for each domain m , in m^2/s , z_i is the charge number of species i (dimensionless), z_i is the charge number of species i , F is Faraday's constant (96485 C mol^{-1}), R is the universal gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$), T is the temperature in K and ϕ_1 is the electrolyte potential in V.

The effective diffusion coefficient (D_i^{eff}) is defined as:

$$D_{i,m}^{eff} = D_i \epsilon_m^b \quad [3]$$

where D_i is the molecular diffusion coefficient of species i , in $m^2 s^{-1}$, b is the Bruggeman correlation factor (assumed $b = 1.5$ - Table IV).

The source term $r_{i,m}$ accounts for the production or consumption of species i due to electrochemical reactions and is defined as:

$$r_{i,m} = a_m \sum_{k=1}^k \frac{\nu_{k,i} i_{k,m}}{F} \quad [4]$$

where a_m is the specific surface area for electrochemical reactions in the cathode (for the separator $a_m = 0$ and implicitly no production of species at the separator), in m^{-1} , $\nu_{k,i}$ is the stoichiometric coefficient of species i in reaction k and $i_{k,m}$ is the interfacial current density of reaction k in which the species i participate in domain m (separator only, when no anode reaction are considered) in $A m^{-2}$.

For the cathode domain, a_m is defined as:

$$a_m = a_{cat} = a_0 \left(\frac{\epsilon}{\epsilon_{cat}} \right)^{1.5} \quad [5]$$

Where a_0 is a reference surface area and ϵ_{cat} is the initial cathode porosity (Table V).

For the anode domain ($m = an$) the reactions are activated only on OCV condition) and $r_{i,k}$ is expressed as a boundary flux with no specific surface area, becoming:

$$r_{i,an} = \sum_{k=1}^k \frac{\nu_{k,i} i_{k,an}}{F} \quad [6]$$

The rate of production or dissolution of precipitates ($R_{p,j,m}$) is given as:

$$R_{p,j,m} = - \sum_j \gamma_{j,k} R_{k,j,m} \quad [7]$$

Table IV. Fundamental constants & physical parameters.

Symbol	Value	Unit	Description
F	96485	$C \text{ mol}^{-1}$	Faraday's constant
R	8.314	$J \text{ mol}^{-1} \text{ K}^{-1}$	Universal gas constant
T	298.15	K	Temperature
α	0.5	-	Charge transfer coefficient
b	1.5	-	Bruggeman exponent

where $\gamma_{j,k}$ is the stoichiometric coefficient of solid species j (Table VI) in reaction k and $R_{k,j,m}$ is the reaction rate for precipitation or dissolution of species i in domain m , in $\text{mol m}^{-3} \text{ s}^{-1}$.

Only Li_2S is modeled as the terminal solid precipitate, in line with its low solubility and thermodynamic stability under open-circuit conditions. While intermediate solid products such as Li_2S_2 have been included in other studies (e.g., Kumaresan et al.¹), their omission here is justified by the focus on equilibrium-driven self-discharge, where Li_2S dominates.

The net reaction rate $R_{k,j,m}$ combines precipitation and dissolution contributions and is expressed as:

$$R_{k,j,m} = k_{j,1} \epsilon_j (c_i^{n_j} - K_j) + k_{j,2} \frac{c_i^{n_j}}{\epsilon_j} \quad [8]$$

$k_{j,1}$ is the precipitation rate constant of species, in s^{-1} , $k_{j,2}$ is the dissolution rate constant of species, in $m^6/(\text{mol}^2 \cdot s)$ or s^{-1} , depending on the mechanism, ϵ_j is the volume fraction of solid species c_1 is the concentration of the limiting species in the electrolyte involved in the precipitation/dissolution of j , n_j is the reaction order with respect to c_1 (e.g., $n_j = 1$ for sulfur, $n_j = 2$ for lithium sulfide), K_j is the solubility limit or solubility product for species j , in $\text{mol}^{n_j}/m^{3n_j}$. Details about the second term can be find in Ref. 2.

It is worth noting that to ensure numerical stability and prevent divergence of the solution when species concentrations approach zero (especially near limiting currents or in depletion zones), a lower-bound cutoff was imposed on all concentration-dependent terms in the model. This numerical treatment replaces the true concentration c_i with:

$$c_i = \max(c_i, 10^{-15}) \quad [9]$$

in any expression where c_i appears as a multiplier or exponent. This practice prevents division by zero or the evaluation of logarithms (e.g. Nernst equation) or powers of near-zero concentrations, which could otherwise lead to numerical instability or non-physical values. The chosen lower threshold value ($10^{-15} \text{ mol m}^{-3}$) is sufficiently small to avoid affecting the physical accuracy of the model while preserving numerical robustness.

Porosity change.—The local porosity ϵ_m within domain m (separator or cathode) evolves dynamically due to the precipitation

Table V. Porosity & conductivity parameters.²

Symbol	Value	Unit	Description
ϵ_{cat}	0.778	-	Initial cathode porosity
ϵ_{sep}	0.37	-	Separator porosity
a_0	132762	m^2/m^3	Reference specific surface area
σ_{cat}	60	S m^{-1}	Electronic conductivity of cathode

The reference active surface area a_0 is adopted from Kumaresan et al.¹ and used as a baseline in the Bruggeman-type scaling of Eq. 5. It is not derived from explicit microstructural modeling in this work.

Table VI. Solid species index j.

Index, j	Species	Notes
1	$\text{S}_{8(\text{s})}$	Solid sulfur (initial cathode material)
2	$\text{Li}_2\text{S}_{(\text{s})}$	Precipitated Li_2S

and dissolution of solid species. The total porosity is defined as the fraction of void space available for electrolyte transport.

The porosity evolution is governed by the following mass balance equation:

$$\frac{\partial \epsilon_m}{\partial t} = - \sum_{j=\text{S}_{8(\text{s})}, \text{Li}_2\text{S}_{(\text{s})}} V_j R_{p,j,m} \quad [10]$$

where V_j is the molar volume of solid species j , in $\text{m}^3 \text{mol}^{-1}$ (Table VII). And the volume fraction of each precipitate is given by:

$$\frac{\partial \epsilon_j}{\partial t} = V_j R_{p,j,m} \quad [11]$$

Boundary conditions.—At the anode boundary ($x = 0$), the species fluxes $\mathbf{N}_{i,\text{an}}$ are imposed based on the interfacial current densities of the Li-metal oxidation and associated polysulfide reduction reactions, assuming a non-porous electrode:

$$\mathbf{N}_{i,\text{an}}|_{x=0} = \sum_{k=1}^k \frac{\nu_{k,i} i_{k,\text{an}}}{F} \quad [12]$$

where $i_{k,\text{an}}$ is the interfacial current density of reaction k at the anode boundary. It is worth noting that the interfacial current densities $i_{k,\text{an}}$ imposed at the anode boundary correspond to “parasitic” reactions primarily associated with self-discharge under open-circuit conditions. These include the slow oxidation of lithium metal and the reduction of diffusing polysulfide species. The reactions are activated only during self-discharge, when $i_{\text{applied}} = 0$. Consequently, the magnitude of $i_{k,\text{an}}$ is significantly lower compared to the interfacial current densities $i_{k,\text{cat}}$ in the cathode domain during typical charge or discharge operation. This difference reflects the

distinct physical regimes modeled: while the cathode is subject to high reaction rates driven by externally applied currents, the anode boundary conditions account for relatively slow self-discharge kinetics occurring spontaneously due to the chemical environment of the Li/S cell. For this reason, the exchange current densities used for the anode reactions are given as:

$$i_{k,\text{an}} = \frac{i_{0,k,\text{cat}}}{0.9^{10}} \quad [13]$$

where $i_{0,k,\text{cat}}$ is the exchange current density associated with the anode reactions, in A m^{-2} .

At the cathode current collector ($x = L_c$), a zero-flux condition is applied for all species:

$$\mathbf{N}_{i,m}|_{x=L_c} = 0 \quad [14]$$

This reflects the assumption that no species can leave or enter the cell through the cathode current collector.

At the interface between the separator and cathode domains, continuity conditions for both concentration and flux are imposed:

$$c_{i,\text{sep}}|_{x=L_s} = c_{i,\text{cat}}|_{x=L_s} \quad [15]$$

$$\mathbf{N}_{i,\text{sep}}|_{x=L_s} = \mathbf{N}_{i,\text{cat}}|_{x=L_s} \quad [16]$$

Charge conservation.—Electrolyte phase.—The current in the electrolyte in the separator and cathode $\mathbf{i}_l$ satisfies:

$$\nabla \cdot \mathbf{i}_{l,m} = -a_m \sum_k i_{k,m} \quad [17]$$

where $\mathbf{i}_{l,m}$ is the ionic current density in the electrolyte in A m^{-2} , with the summation over all charged species:

$$\mathbf{i}_{l,m} = F \sum_i z_i \mathbf{N}_{i,m} \quad [18]$$

where $i_{k,m}$ is the interfacial current density of reaction k occurring at the solid-liquid interface in domain m , in A m^{-2} . In the separator domain $m = \text{sep}$, $a_{\text{sep}} = 0$ since no electrochemical reactions occur.

Solid phase (cathode only).—In the porous solid phase (cathode), the electronic current $\mathbf{i}_s$ is governed by:

$$\nabla \cdot \mathbf{i}_{s,m} = -a_m \sum_k i_{k,m} \quad [19]$$

where $\mathbf{i}_{s,m}$ is the electronic current density, in A m^{-2} defined only in the cathode as:

$$\mathbf{i}_{s,m} = -\sigma_{\text{eff},m} \nabla \phi_{s,m} \quad [20]$$

where $\sigma_{\text{eff},m}$ is the effective electrical conductivity of the cathode material, in S m^{-1} , defined as:

Table VII. Molar volumes and rate constants used for precipitation and dissolution reactions of solid sulfur and lithium sulfide.²

Symbol	Value ^{a)}	Unit	Description
$V_{\text{S}_{8(\text{s})}}$	$1.239 \cdot 10^{-4}$	$\text{m}^3 \text{mol}^{-1}$	Molar volume of sulfur solid
$V_{\text{Li}_2\text{S}_{(\text{s})}}$	$2.768 \cdot 10^{-5}$	$\text{m}^3 \text{mol}^{-1}$	Molar volume of Li_2S solid
$k_{\text{S}_{8(\text{s})}}$	100	s^{-1}	Precipitation rate constant for $\text{S}_{8(\text{s})}$
$k_{\text{S}_{8(\text{s}),2}}$	$1 \cdot 10^{-2}$	s^{-1}	Dissolution rate constant for $\text{S}_{8(\text{s})}$
$K_{\text{S}_{8(\text{s})}}$	19	mol m^{-3}	Solubility limit of $\text{S}_{8(\text{s})}$ in electrolyte
$k_{\text{Li}_2\text{S}_{(\text{s})}}$	$1.81 \cdot 10^{-6}$	$\text{m}^6 \text{s}^{-1} \text{mol}^{-2}$	Precipitation rate constant for $\text{Li}_2\text{S}_{(\text{s})}$
$k_{\text{Li}_2\text{S}_{(\text{s}),2}}$	$1 \cdot 10^{-16}$	$\text{m}^6 \text{s}^{-1} \text{mol}^{-2}$	Dissolution rate constant for $\text{S}_{8(\text{s})}$
$K_{\text{Li}_2\text{S}_{(\text{s}),2}}$	$1 \cdot 10^7$	$\text{mol}^3 \text{m}^{-9}$	Solubility limit of $\text{Li}_2\text{S}_{(\text{s})}$ in electrolyte

a) Values taken from Ref. 2, with some changes to emphasize the voltage profile characteristic to Li/S.

Table VIII. List of reactions at the different domains.

Reaction index, k	Reaction	Domain	Notes
During Discharge and OCV			
1	$\text{Li} \rightleftharpoons \text{Li}^+ + \text{e}^-$	Anode	Main discharge reaction at anode
2–6	$\text{S}_8 \rightleftharpoons \text{S}_8^{2-} \rightleftharpoons \text{S}_6^{2-} \rightleftharpoons \text{S}_4^{2-} \rightleftharpoons \text{S}_2^{2-} \rightleftharpoons \text{S}_0^{2-}$	Cathode	Main sulfur reduction/oxidation steps
During OCV Only ^{a)}			
7–11	$\text{S}_8 \rightleftharpoons \text{S}_8^{2-} \rightleftharpoons \dots \rightleftharpoons \text{S}_0^{2-}$	Anode Boundary	Parasitic self-discharge reactions at anode boundary

a) Although the reactions at the anode are the same as at the cathode, they are denoted with 7–11 to differentiate them from the cathode, because they have different values for $i_{0,k,\text{ref}}$ and the Butler-Volmer formulations.

$$\sigma_{\text{eff},m} = \sigma_m(1 - \epsilon_m)^b \quad [21]$$

where ϕ_s is the solid potential in the cathode, in V, σ_m is the electrical conductivity of the cathode material. In the separator domain, there is no electronic conduction, i.e., $\sigma_{\text{sep}} = 0$.

Boundary conditions.—At the anode boundary ($x = 0$):

$$i_{l,\text{an}}|_{x=0} = i_{k,0} \quad [22]$$

when no anode reactions and:

$$i_{l,\text{an}}|_{x=0} = \sum i_{k,0} \quad [23]$$

where adding the anode reactions. There is no solid-phase current at the anode.

At the separator/cathode boundary ($x = L_s$) there is no flux:

$$i_{s,\text{sep}}|_{x=L_s} = 0 \quad [24]$$

$$i_{l,\text{sep}}|_{x=L_s} = i_{l,\text{cat}}|_{x=L_s} \quad [25]$$

At the cathode boundary a current density is applied:

$$i_{s,\text{cat}}|_{x=L_c} = i_{\text{applied}} \quad [26]$$

Kinetics

Each step of the sulfur reduction in the cathode ($m = \text{cat}$) follows Butler-Volmer kinetics:

$$i_{k,m} = i_{0,k} \left[\left(\frac{c_{i,R}}{c_{R,i,\text{ref}}} \right)^{\nu_{R,k}} \exp\left(\frac{\alpha_a F \eta_k}{RT}\right) - \left(\frac{c_{i,O}}{c_{O,i,\text{ref}}} \right)^{-\nu_{O,k}} \exp\left(-\frac{\alpha_c F \eta_k}{RT}\right) \right] \quad [27]$$

where $i_{k,\text{cat}}$ is the reaction current density for reaction k at the cathode in A m^{-2} , $i_{0,k}$ is the exchange current density in A m^{-2} , $c_{i,\text{ref}}$ is the reference concentrations of each i species participating in reaction k (Table VIII), in mol m^{-3} , α is the charge transfer coefficient (dimensionless, assumed $\alpha = \alpha_a = \alpha_c = 0.5$), R and O are the index corresponding to the reduction side of the reaction and the oxidation side, respectively, η_k is the overpotential in V defined as:

$$\eta_{k,m} = \phi_s - \phi_{l,m} - U_{k,\text{ref}} \quad [28]$$

where $U_{k,\text{ref}}$ is the equilibrium potential for reaction k in V, calculated as:

$$U_{k,\text{ref}} = U_{0,k} - \frac{RT}{F} \left(\nu_{i,R} \ln\left(\frac{c_{R,\text{ref}}}{1000}\right) + \nu_{i,O} \ln\left(\frac{c_{O,\text{ref}}}{1000}\right) \right) \quad [29]$$

A unique contribution to the paper is the addition of the reactions occurring at the anode surface, which, in the model are added to a boundary. Adding a full Butler-Volmer kinetics formulation adds high non-linearities which are hard to deal with, even with strong non-linear solvers and high-density mesh. For this proof of concept and to reduce the computational burden and eliminate any numerical noise, the kinetics at the anode are modeled using a linearized Butler-Volmer formulation. Considering that the kinetics during open circuit at the anode are sluggish compared to the operation kinetics, such as assumption is physically valid. Therefore, at the anode, the linearized Butler-Volmer kinetics are defined as:

$$i_{k,\text{an}} = i_{0,k,\text{an}} \left[\frac{(\alpha_c + \alpha_a)F}{RT} \right] \cdot \eta_{k,\text{an}} \quad [30]$$

where the overpotential $\eta_{k,\text{an}}$ is defined as:

$$\eta_{k,\text{an}} = \phi_l - U_{k,\text{ref}} \quad [31]$$

The reactions modeled are summarized in Table IX.

Initial conditions.—The initial concentrations used in the model are summarized in Table X.

Results

Experimental data.—To complement the modeling results, experimental data from Wen et al.²⁸ was used to qualitatively illustrate the characteristic voltage decay associated with self-discharge in Li/S cells. Figure 4 shows the experimentally measured OCV decay over time for a Li/S cell starting at a cutoff voltage of ~ 2.18 V, after a partial discharge.

As seen in Fig. 4, the voltage decays continuously over approximately 48 h, dropping from an initial OCV of 2.18 V to about 2.15 V. This behavior is consistent with the parasitic reduction of dissolved polysulfides at the lithium-metal anode.

It is important to note that this experimental dataset is used here solely for qualitative validation of the model's physical assumptions regarding self-discharge behavior. No fitting or parameter calibration is performed against these results, which remains beyond the scope of this study. The experimental conditions, electrode architecture, and electrolyte formulation used in the work of Wen et al. differ from those assumed in the present modeling framework. Nevertheless, the observed voltage decay trends are representative of the general self-discharge behavior expected in Li/S cells.

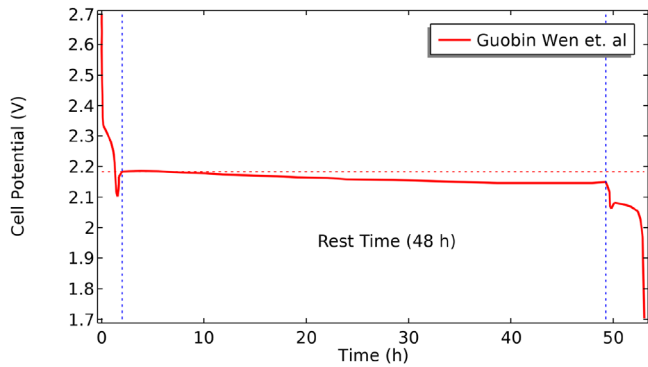
Modeling data (full discharge—no anode reactions).—The voltage calculated using the Li/S battery model is shown in Fig. 5. The figure shows the typical voltage profile corresponding to a Li/S chemistry during discharge. The cell exhibits the well-known two-plateau behavior. The first plateau initiates at an open-circuit voltage of approximately 2.45 V and stabilizes near 2.35–2.38 V for the first 10 h corresponding to the reduction of elemental sulfur $\text{S}_{8(l)}$ into

Table IX. List of reactions and the parameters.^{1,2}

Reaction <i>k</i>		Reaction	$i_{0,k} \text{ A m}^{-2}$	$U_k \text{ V}$	ν
cat	an				
	1	$\text{Li} \rightarrow \text{Li}^+ + e^-$	0.394	0	$\nu_{\text{Li}} = -1$
2	7	$0.5\text{S}_{8(l)} + e^- \rightarrow 0.5 \text{S}_8^{2-}$	1.972	2.39	$\nu_{2,\text{S}8} = -1/2$
3	8	$1.5\text{S}_8^{2-} + e^- \rightarrow 2\text{S}_6^{2-}$	0.019	2.37	$\nu_{2,\text{S}82} = 1/2$ $\nu_{3,\text{S}82} = -3/2$
4	9	$\text{S}_6^{2-} + e^- \rightarrow 1.5\text{S}_4^{2-}$	0.019	2.24	$\nu_{3,\text{S}62} = 2$ $\nu_{4,\text{S}62} = -1$
5	10	$0.5\text{S}_4^{2-} + e^- \rightarrow \text{S}_2^{2-}$	1.97e-4	2.04	$\nu_{4,\text{S}42} = 3/2$ $\nu_{5,\text{S}42} = -1/2$
6	11	$0.5\text{S}_2^{2-} + e^- \rightarrow \text{S}^{2-}$	1.97e-7	2.11	$\nu_{5,\text{S}22} = 3/2$ $\nu_{6,\text{S}42} = -1/2$ $\nu_{5,\text{S}2} = 1$

Table X. Initial conditions and Diffusion Coefficients.

Species, <i>i</i>	$c_{i,\text{ref}} \text{ mol m}^{-3}$	$D_0 \text{ m}^2 \text{ s}^{-1}$
$\text{S}_{8(l)}$	19	$1 \cdot 10^{-10}$
S_8^{2-}	0.178	$6 \cdot 10^{-10}$
S_6^{2-}	0.324	$6 \cdot 10^{-10}$
S_4^{2-}	0.02	$1 \cdot 10^{-10}$
S_2^{2-}	$5.229 \cdot 10^{-7}$	$1 \cdot 10^{-10}$
S^{2-}	$8.267 \cdot 10^{-10}$	$1 \cdot 10^{-10}$
Li^+	1001	$1 \cdot 10^{-10}$
A^-	1000	$4 \cdot 10^{-10}$

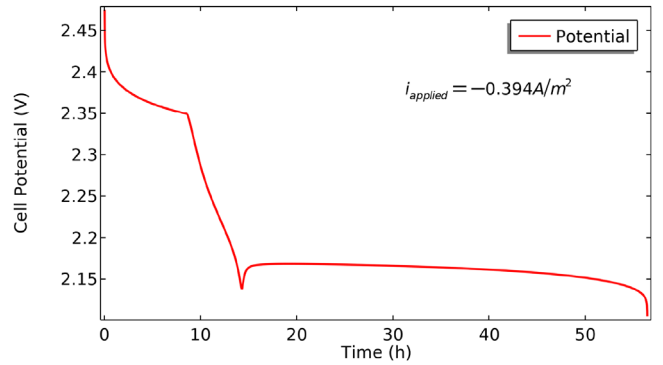
**Figure 4.** Experimentally measured open-circuit voltage (OCV) decay of a Li/S cell during rest after partial discharge, adapted from Wen et al.²⁸

higher-order polysulfide species such as S_8^{2-} and S_6^{2-} (see Fig. 6). As sulfur is consumed in the cathode, the cell transitions into a second plateau which persists for the remainder of the discharge until about 53 h. This region corresponds to the stepwise reduction of lower-order polysulfides S_4^{2-} and S_2^{2-} , and eventually S^{2-} , leading to the precipitation of solid $\text{Li}_2\text{S}_{(s)}$ (see Fig. 7). The applied current density for this simulation is -0.394 A m^{-2} , leading to a complete discharge and voltage drop near 2.12 V.

In Fig. 6 one can see the evolution of the spatially average concentrations of all dissolved species over time, defined as:

$$c_{i,\text{avg}} = \frac{1}{L - L_c} \int_{x=L}^{x=L_c} c_i dx \quad [32]$$

Initially, the concentration of $\text{S}_{8(l)}$ decreases rapidly as it is reduced at the cathode surface, while the concentration of higher

**Figure 5.** Discharge voltage profile of a Li/S cell under galvanostatic conditions ($i_{\text{applied}} = -0.394 \text{ A m}^{-2}$) with the anode modeled as a boundary and no anode reactions included.

order polysulfides (S_8^{2-} , S_6^{2-} , S_4^{2-}) increases accordingly. These species are then consumed in subsequent reduction steps, leading to the transient appearance and disappearance of intermediate polysulfides, as seen from the multiple crossovers in the concentration profiles. As the discharge progresses, lower-order polysulfides (S_2^{2-}) start to accumulate, before their eventual reduction to S^{2-} . Toward the end of the discharge, the concentration of S^{2-} reaches a maximum, and the system becomes dominated by the precipitation of $\text{Li}_2\text{S}_{(s)}$, as the polysulfide species are depleted.

Figure 7 illustrates the evolution of the volume fractions of solid precipitates, $\text{S}_{8(s)}$ and $\text{Li}_2\text{S}_{(s)}$, across the different boundaries of the cell (cathode, separator, and anode boundary). During the initial stages of discharge, $\text{S}_{8(s)}$ precipitation occurs mainly within the cathode due to the local supersaturation of dissolved sulfur species. However, as discharge proceeds and sulfur is consumed, $\text{S}_{8(s)}$ dissolves, and $\text{Li}_2\text{S}_{(s)}$ precipitation becomes dominant, especially during the second plateau region. The volume fraction of $\text{Li}_2\text{S}_{(s)}$ increases steadily within the cathode, eventually occupying a significant portion of the pore volume.

The porosity ε_m in the cathode decreases in response to the accumulation of solid precipitates. This effect is particularly pronounced in the cathode, where the highest rate of electrochemical reactions occurs. The decrease in porosity directly impacts mass transport properties, contributing to the gradual voltage decay observed near the end of discharge. In contrast, only minor precipitation of $\text{S}_{8(s)}$ and $\text{Li}_2\text{S}_{(s)}$ is observed in the separator and at the anode boundary, reflecting the lower local concentrations of polysulfide species in these regions.

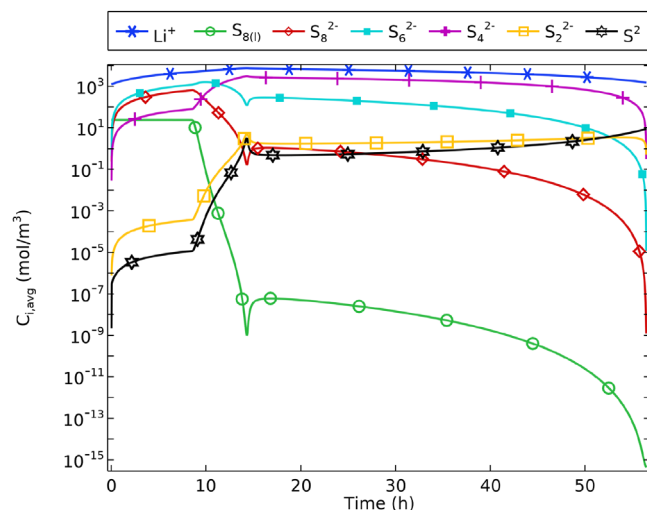


Figure 6. Evolution of spatially averaged concentrations of dissolved species during discharge without anode-side reactions.

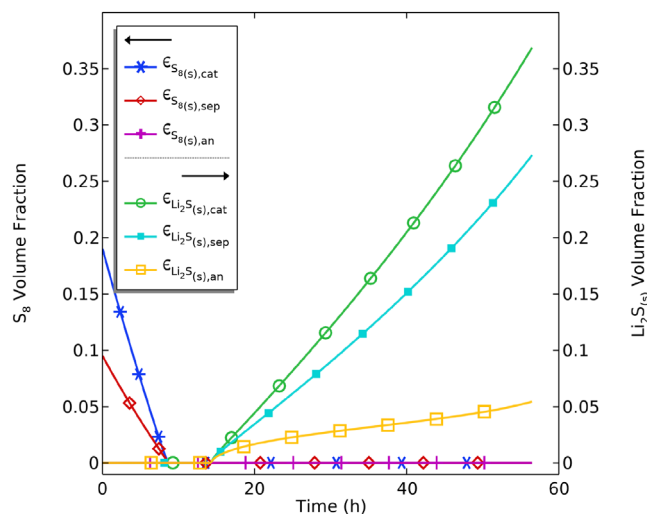


Figure 7. Volume fraction evolution of solid sulfur $S_{8(s)}$ and $Li_2S_{(s)}$ precipitates within the cathode, separator, and anode regions during discharge. The cathode shows significant $Li_2S_{(s)}$ accumulation, reducing porosity and limiting transport at the end of discharge.

Conditions at open circuit (with no anode reactions).—To further analyze the behavior of the Li/S cell in the absence of parasitic reactions, an open-circuit simulation was performed following a short discharge pulse. In this scenario, the cell was initially discharged under a constant current density of -0.394 A m^{-2} until a cutoff voltage of 2.35 V, and the system was left to evolve under open-circuit conditions for approximately 52 h. After the cutoff voltage, the voltage exhibits a voltage recovery to about 2.36 V, where it stays constant for the entire 52 h (Fig. 8).

During the initial discharge phase, the voltage drops rapidly from as the reduction of $S_{8(l)}$ proceeds and higher-order polysulfide species begin to form. However, because the discharge period is short (lasting approximately 8.3 h), the system does not progress into the regime where lower-order polysulfides or solid $Li_2S_{(s)}$ are formed in significant quantities (Fig. 10). This behavior reflects a thermodynamic equilibrium governed by the dominant presence of higher-order polysulfides, such as S_8^{2-} , S_6^{2-} , S_4^{2-} , which remain dissolved in the electrolyte. The small difference between the cutoff and final stabilized voltage indicates that in the absence of anode reactions or further electrochemical activity, the system reaches a quasi-equilibrium state immediately after the current is removed.

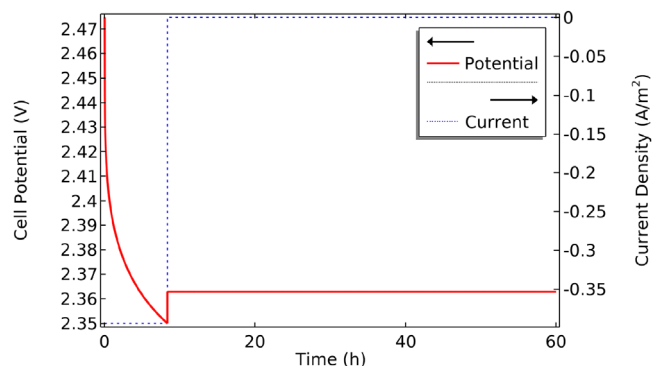


Figure 8. Simulated voltage and current profile during a short discharge pulse at $i_{\text{applied}} = -0.394 \text{ A m}^{-2}$ at a cutoff voltage of 2.35 V, followed by open-circuit conditions without anode-side reactions.

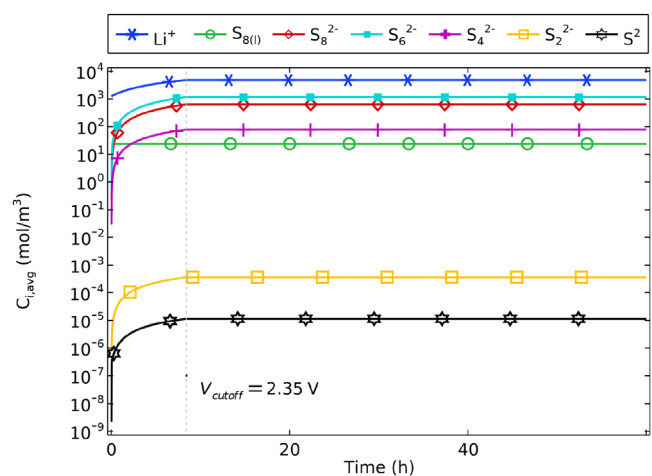


Figure 9. Evolution of average concentrations of dissolved species during partial discharge and subsequent open-circuit relaxation without anode reactions.

The evolution of species concentrations during this process is presented in Fig. 9.

Figure 9 shows the evolution of species concentrations during this process. As expected, during the short discharge period, the concentration of $S_{8(l)}$ decreases steadily, while the concentrations of higher-order polysulfides, such as S_8^{2-} and S_6^{2-} , increase as intermediate reduction products. However, it is important to note that the consumption of $S_{8(l)}$ is relatively slow, as indicated by its gradual decrease even after several hours of discharge. This slow depletion is attributed to the limited applied current density, which imposes a mild driving force for sulfur reduction, combined with the low solubility and relatively slow dissolution kinetics of $S_{8(l)}$ into the electrolyte. As a result, the reduction of $S_{8(l)}$ is not transport-limited, but rather reaction-limited, occurring predominantly at the cathode-electrolyte interface where dissolved sulfur is available.

Lower-order polysulfides, such as S_4^{2-} , S_2^{2-} , and S^{2-} , remain at very low concentrations throughout the discharge, consistent with the shallow depth of discharge and the early interruption of the current. Once the applied current is interrupted, all species concentrations stabilize and remain constant over the remainder of the simulation. This behavior confirms the absence of further electrochemical reactions or species redistribution in the system, as no parasitic anode reactions are included in this configuration.

Figure 10 illustrates the evolution of the solid-phase volume fractions during this process. In this simulation, anode-side reactions are excluded, and the cell is only partially discharged. The volume fraction of $S_{8(s)}$ is observed to decrease steadily, indicating that solid

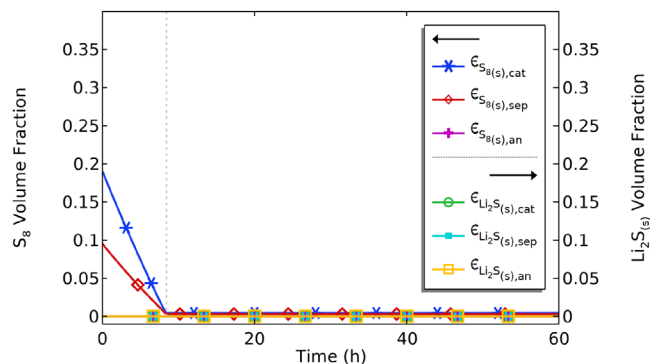


Figure 10. Volume fraction evolution of solid sulfur $S_{8(s)}$ precipitates within the cathode, separator, and anode during partial discharge without anode reactions.

sulfur is undergoing dissolution as it is reduced to soluble polysulfide species. Since the depth of discharge is limited and no anode-side reactions are active, the system never exceeds the solubility limit of sulfur. As a result, no additional precipitation occurs during this stage. Once the current is removed, species concentrations stabilize, and the amount of remaining solid sulfur remains constant. Notably, no $Li_2S_{(s)}$ formation is observed in this scenario, as the discharge was interrupted before lower-order polysulfide formation could proceed to the point of precipitation.

Figure 11 illustrates the evolution of species concentrations throughout the entire cell during discharge and OCV, including the anode, separator, and cathode regions. In this configuration, the model excludes all anode-side reactions, allowing the system to evolve solely based on the electrochemical processes occurring in the cathode and the transport of ionic species through the electrolyte. From Fig. 11 it can be seen that the concentrations vary during discharge until the cutoff voltage (indicated in the figures with dashed vertical grey lines). It can be seen that lithium starts to be produced, followed immediately by a decrease in the concentration of $S_{8(l)}$, then an increase in the low polysulfide concentrations. However, once the current is set to zero, the concentration remains constant for the entire duration of the open circuit. A notable feature observed across all plots is that the concentration profiles for every species overlap across all domains. That is, the values of Li^+ , $S_{8(l)}$, all intermediate polysulfides (S_8^{2-} , S_6^{2-} , S_4^{2-} , S_2^{2-} , S^{2-}), and the counterion A^- are identical in the anode, separator, and cathode regions throughout the discharge. This behavior indicates that the system reaches a spatially homogeneous state very rapidly, with no significant gradients in species concentrations forming over time.

The uniformity arises from the combination of high diffusion coefficients, a relatively short cell geometry in the 1D model, and the absence of reactions outside the cathode domain. As soon as discharge begins, lithium ions generated at the anode boundary diffuse rapidly through the separator and into the cathode, resulting in a uniform increase in lithium concentration throughout the cell. The reduction of $S_{8(l)}$ leads to the formation of higher-order polysulfides such as S_8^{2-} and S_6^{2-} , which also equilibrate quickly across the entire electrolyte volume. The concentrations of these intermediates rise sharply in the early stages of discharge and then stabilize, forming a quasi-steady-state profile. In contrast, the lower-order species S_4^{2-} , S_2^{2-} , and S^{2-} remain at extremely low concentrations throughout the simulation, consistent with the high cutoff voltage of 2.35 V that limits the extent of sulfur reduction. The anion concentration decreases uniformly in response to the buildup of Li^+ , maintaining electroneutrality throughout the domains.

This configuration serves as a well-defined baseline for understanding cathode-limited behavior, but also highlights that in a real system, additional processes - particularly parasitic reactions at the anode - are expected to drive continued cell degradation or voltage fade even under open-circuit conditions.

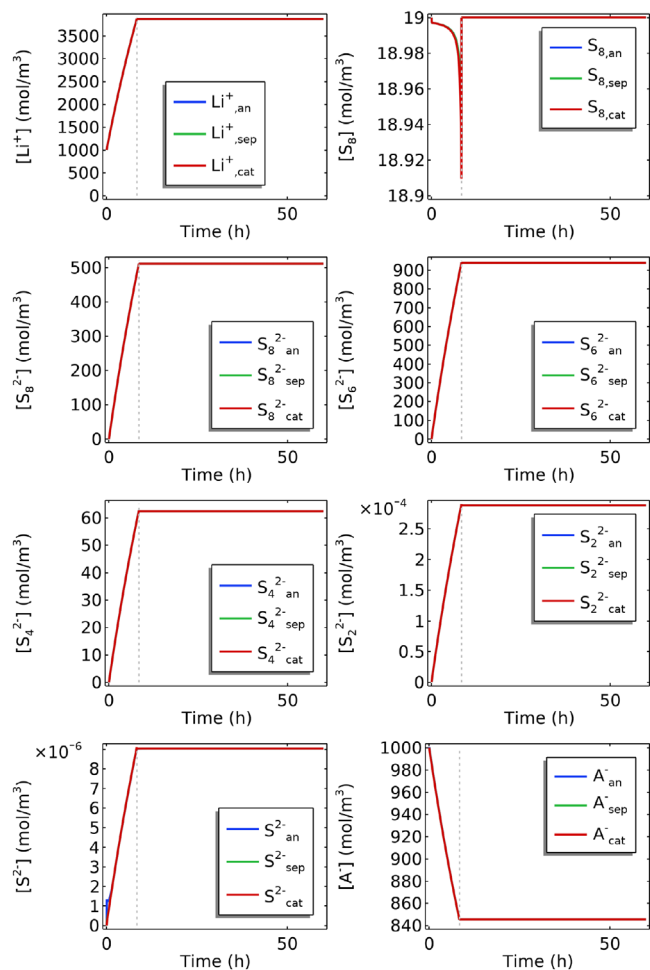


Figure 11. Species concentrations in the cathode, separator, and anode regions during shallow discharge and open circuit of a Li/S cell for the case with no anode reactions.

Conditions at open circuit (with anode reactions).—To assess the role of self-discharge and anode-side parasitic reactions, the Li/S cell model was extended to include lithium-metal oxidation and polysulfide reduction at the anode boundary (see Fig. 1b). The simulation conditions were otherwise identical to the previous case, that is, the cell was discharged at a constant current density of -0.394 A m^{-2} to a cutoff voltage of 2.30 V, followed by an open-circuit rest period.

Figure 12 shows the voltage and current evolution during this scenario. As before, the voltage drops from an initial open-circuit value of 2.47 V during the discharge period until the cutoff voltage. However, once the current is interrupted and the cell is left at open circuit, a voltage decay is observed over time due to the inclusion of anode-side reactions. The voltage decays gradually from 2.31 V down to approximately 2.28 V over the next 52 h. This behavior aligns qualitatively with the experimental data presented earlier (Fig. 4), confirming the essential role of parasitic reactions in driving voltage loss even in the absence of external current flow.

From Fig. 13 it shows a similar trend with the previous case (with no anode reactions), which indicates that the changes are slow, which is expected, considering the sluggish behavior of the parasitic reactions at the anode. However, when looking at the concentration of each species (Fig. 14), it can be seen that during open circuit, the reactions are active. For example, after the discharge is interrupted at around 8.39 h, the Li^+ concentration increases slightly, while the concentration of $S_{8(l)}$ shows a slow but steady decline across all domains, including the separator and anode side - despite the absence of an external applied current. This indicates that $S_{8(l)}$ continues to be consumed, albeit at a much slower rate, due to

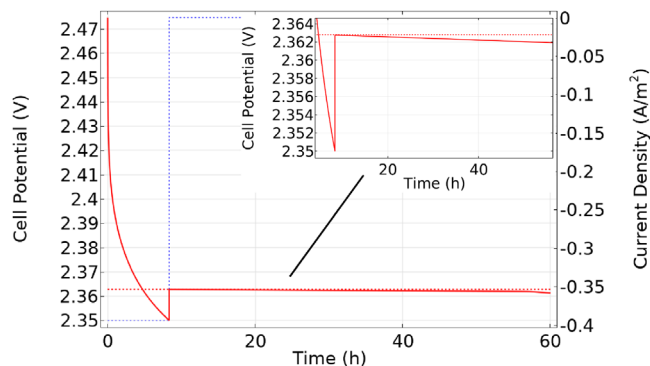


Figure 12. Simulated voltage and current profile during a short discharge pulse at $i_{\text{applied}} = -0.394 \text{ A m}^{-2}$ at a cutoff voltage of 2.35 V, followed by open-circuit conditions with parasitic reactions added at the anode-side.

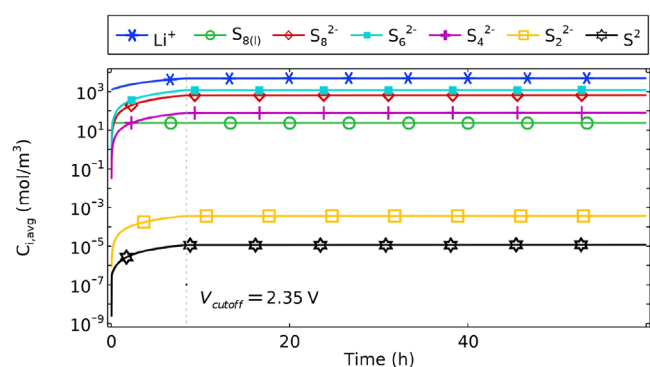


Figure 13. Evolution of average concentrations of dissolved species during partial discharge and subsequent open-circuit relaxation with anode reactions.

ongoing reduction at the anode interface. Simultaneously, the S_8^{2-} , S_6^{2-} , and S_4^{2-} species exhibit a slight upward trend during open circuit, reflecting the continuation of the sulfur reduction cascade. The emergence and growth of S_2^{2-} and S^{2-} , though still at low concentrations, further supports the occurrence of multi-step electrochemical reduction processes even in the absence of external current. Additionally, the continued decline in A^- concentration across the entire rest period confirms sustained ionic movement to maintain charge neutrality as electrons are transferred at the anode.

Figure 15 illustrates the evolution of the solid-phase volume fractions for precipitated sulfur ($\text{S}_{8(s)}$) and lithium sulfide ($\text{Li}_2\text{S}_{(s)}$) across the cathode, separator, and anode domains during a discharge pulse followed by open-circuit relaxation with parasitic reactions at the anode.

During the initial discharge, the volume fraction of $\text{S}_{8(s)}$ drops sharply, particularly in the cathode and separator, indicating active dissolution of the solid sulfur phase as it is electrochemically reduced to soluble polysulfides. This trend continues more gradually during open-circuit rest, driven by slow parasitic reduction of remaining dissolved sulfur species at the anode, as evidenced by the continued but slower decline in ϵ_{S_8} in all domains. The volume fraction of $\text{S}_{8(s)}$ in the anode remains relatively low throughout, but nonzero, highlighting the migration of sulfur species and possible redeposition via chemical equilibria or back-diffusion.

In parallel, $\text{Li}_2\text{S}_{(s)}$ volume fractions rise slightly in all three domains. Although the overall values remain low (on the order of 10^{-7}), the appearance of solid $\text{Li}_2\text{S}_{(s)}$ confirms that parasitic reactions continue to reduce lower-order polysulfides even during open circuit, leading to gradual precipitation. The concentration in the cathode stays minimal, consistent with the fact that $\text{Li}_2\text{S}_{(s)}$ forms primarily where reduction of soluble species meets free Li^+

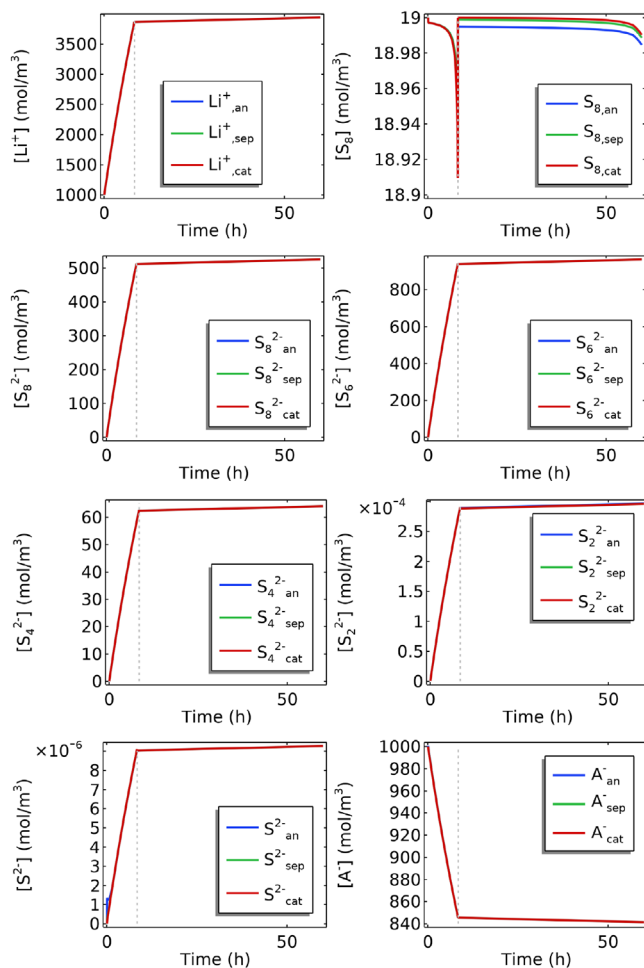


Figure 14. Species concentrations in the cathode, separator, and anode regions during shallow discharge and open circuit of a Li/S cell for the case with anode electrochemical reactions.

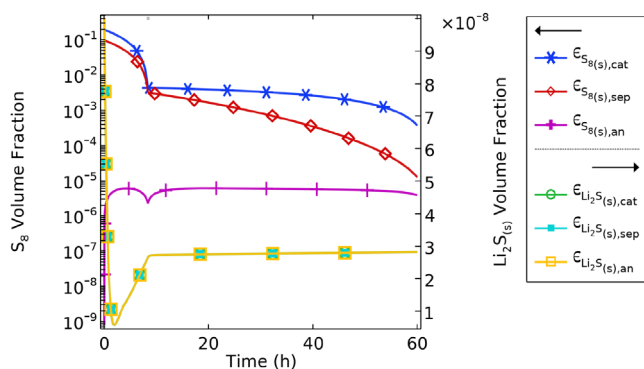


Figure 15. Volume fraction evolution of solid sulfur $\text{S}_{8(s)}$ precipitates within the cathode, separator, and anode regions during partial discharge with anode reactions.

availability, offering conditions more readily satisfied near the lithium interface.

Discussion

In most prior modeling efforts, self-discharge in Li/S batteries has been described in terms of shuttling or chemical redox reactions at the anode. These treatments rely on phenomenological or lumped-parameter models to represent parasitic loss without explicitly modeling the electrochemical nature of the process.

However, this study demonstrates that electrochemical anode-side reactions are a necessary condition for self-discharge to occur. When no anode reactions are present, the system reaches equilibrium almost instantly after current interruption, and all concentrations remain constant. In contrast, when electrochemical kinetics are added at the anode, the system shows voltage fade, sustained Li^+ production, continued polysulfide conversion, and $\text{Li}_2\text{S}_{(s)}$ precipitation.

This suggests that shuttling alone (i.e., transport of polysulfides without interfacial charge transfer) cannot account for capacity fade under OCV. Instead, Faradaic reactions at the lithium surface - specifically the oxidation of lithium metal and electrochemical reduction of polysulfides - must be present for the system to continue evolving during rest.

The modeling results align qualitatively with experimental data from Wen et al.²⁸ reinforcing the physical relevance of the proposed mechanism. The voltage decay trends, species evolution, and onset of $\text{Li}_2\text{S}_{(s)}$ precipitation all support the idea that self-discharge in Li/S batteries is not merely a transport-driven phenomenon, but a reaction-limited, electrochemical process centered at the lithium anode.

Conclusions

This study presents a new electrochemical modeling framework that explicitly captures the anode-side parasitic reactions responsible for self-discharge in Li/S batteries. Unlike prior models that treat self-discharge either as a purely chemical phenomenon or a consequence of mass transport (i.e., polysulfide shuttling), this work introduces a mechanistically consistent and electrochemically formulated approach based on lithium oxidation and sequential polysulfide reduction at the anode.

Our simulations demonstrate a clear distinction between scenarios with and without anode-side reactions. In the absence of these reactions, the system stabilizes upon current interruption, with species concentrations and voltage remaining constant. This corresponds to a cathode-limited system in which no self-discharge is observed. However, when electrochemical reactions are included at the anode, a slow voltage decay ensues, consistent with experimental observations. The continued formation of lower-order polysulfides, increase in lithium-ion concentration, and precipitation of $\text{Li}_2\text{S}_{(s)}$ all confirm the sustained electrochemical activity at the anode during open-circuit rest. These results support the notion that self-discharge cannot occur without Faradaic reactions at the lithium interface.

Critically, this work challenges the conventional view of the polysulfide shuttle as a transport-limited process and instead reframes self-discharge as an electrochemical issue requiring redox coupling at the anode. The data reveals that migration of polysulfides is not inherently parasitic unless accompanied by electron transfer reactions. Voltage fade and active lithium loss are direct consequences of these reactions, which had previously been oversimplified or neglected.

The model also highlights the subtle interplay between mass transport, reaction kinetics, and precipitation dynamics in governing long-term voltage stability. Precipitate volume fractions, particularly $\text{Li}_2\text{S}_{(s)}$, were found to increase in the separator and anode during rest, emphasizing the distributed nature of parasitic deposition. Moreover, the agreement between our model and experimental OCV decay trends lends confidence to the physical assumptions made, despite the lack of direct calibration.

This work provides a foundational shift in the way self-discharge in Li/S cells is understood and modeled. By resolving anode-side reactions electrochemically, it opens the door to more accurate predictions of capacity fade and shelf-life degradation. Future efforts can build on this framework to explore mitigation strategies - such as anode coatings, artificial SEIs, or selective electrolytes - that specifically target the interfacial reaction pathways identified herein. This model can also be extended to account for full-domain lithium modeling and spatially resolved SEI growth, offering a comprehensive tool for improving Li/S battery design and reliability. In future work, extending the present model to simulate constant current and constant potential charging while incorporating polysulfide reactions

at the anode would provide a quantitative framework for evaluating current efficiency during charge. This would enable predictive insight into coulombic losses and inform strategies for improving charge recovery in practical Li/S cells.

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